

Correction Torsors for Support-Defect Repairs

Autonomous discussion Pass 107 models support-defect repairs as torsors and separates ordinary choice data from Rosser/cosheaf phantom data.

For finite support S , define

$$K_S = \ker \Sigma_S = \{a \in \mathbb{Z}^S : \sum_{p \in S} a_p = 0\}.$$

If coordinate deletion gives $u = d|_S$ with additive defect $\Delta = \Sigma_S(u)$, an additive repair is

$$r = u - a, \quad \Sigma_S(a) = \Delta.$$

Any two additive repairs differ by an element of K_S , and adding an element of K_S to one repair gives another. Thus additive repairs form a free transitive K_S torsor.

Primitive repairs are a refinement, not the whole torsor. On support $\{2, 3\}$, the vector $(1, -1)$ is primitive, but adding the kernel element $(1, -1)$ gives $(2, -2)$, which is zero-sum but nonprimitive. Hence the primitive repair locus is not stable under the full K_S action.

A basepoint $b \in S$ gives a splitting

$$s_b(n) = ne_b.$$

Transitions between basepoints are

$$\tau_{a,b} = s_b(1) - s_a(1) \in K_S,$$

and satisfy the cocycle identity

$$\tau_{a,b} + \tau_{b,c} = \tau_{a,c}.$$

These transitions are coboundaries of chosen splittings. Along support inclusions, basepointed splittings are natural exactly when the basepoint is preserved.

Therefore the support-defect repair data is ordinary finite-level choice data, not a Rosser/cosheaf phantom. The finite exact sequence

$$0 \rightarrow K_S \rightarrow \mathbb{Z}^S \xrightarrow{\Sigma_S} \mathbb{Z} \rightarrow 0$$

splits after choosing a basepoint, and no non-Mittag-Leffler inverse tower is present. Linear repairs commute with the antipode, so the $B\mathbb{Z}/2$ boundary-line local system is unaffected.

The checker `artifacts/reports/pass107-correction-torsors-support-defect-check.json` verifies the additive torsor property, primitive-locus non-stability, basepoint transition coboundaries, inclusion naturality under preserved basepoints, antipode compatibility, and the ordinary-choice-not-phantom verdict.

The next task is to classify the integral equivariant obstruction: rational support-symmetric sections of Σ_S exist by barycenters, but integral support-symmetric sections do not.